

# Use Case Template

## CardioAI - Heart Disease Prediction System

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| <b>1. Use Case Title</b>       | Compute SHAP Values                                                                                                                                                                                                                                                                                                                                                                              |
| <b>2. Abbreviated Title</b>    | Compute SHAP                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>3. Use Case Id</b>          | UC-13                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>4. Actors</b>               | ML Engine (Supporting)                                                                                                                                                                                                                                                                                                                                                                           |
| <b>5. Description</b>          | The ML Engine computes SHAP (SHapley Additive exPlanations) values to quantify each of the 13 clinical features' marginal contribution to the model's prediction output.                                                                                                                                                                                                                         |
| <b>5.1. Pre Conditions</b>     | <ol style="list-style-type: none"><li>1. Model inference (UC-11) has been completed.</li><li>2. Background dataset (training sample) is available for SHAP explainer.</li></ol>                                                                                                                                                                                                                  |
| <b>5.2. Task Sequence</b>      | <ol style="list-style-type: none"><li>1. System initializes DeepSHAP explainer with background data.</li><li>2. System computes SHAP values for the current patient input.</li><li>3. System ranks features by absolute SHAP value.</li><li>4. System generates positive/negative contribution indicators.</li><li>5. System outputs sorted feature importance list for visualization.</li></ol> |
| <b>5.3. Post Conditions</b>    | <ol style="list-style-type: none"><li>1. A vector of 13 SHAP values (one per feature) is produced.</li><li>2. Feature importance ranking is ready for bar/waterfall chart rendering.</li></ol>                                                                                                                                                                                                   |
| <b>6. Modification History</b> | Date: September 11, 2026                                                                                                                                                                                                                                                                                                                                                                         |
| <b>7. Author</b>               | Dhruv Singla (1024170434), Srishti (1024170451)                                                                                                                                                                                                                                                                                                                                                  |